SUSHANT BANJARA

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EDUCATION

The University of Chicago **2021 - 2027**
PhD, Public Policy
Dissertation: Essays on Political Economy of Conservation and Land Use Decisions

Yale University **2017 - 2019**
Master's, Environmental Science
Thesis: Mapping Burned Areas in the Himalayan Region Using Time Series of Landsat Images

Tribhuvan University, Institute of Engineering **2011 - 2016**
Bachelor's, Civil Engineering

FIELDS

Environmental Economics, Political Economy, Development Economics

REFERENCES

Eyal Frank (Chair)
Assistant Professor
Harris School of Public Policy
University of Chicago
eyalfrank@uchicago.edu

Oeindrila Dube
Philip K. Pearson Professor
Harris School of Public Policy
University of Chicago
odube@uchicago.edu

James Robinson
University Professor
Harris School of Public Policy
University of Chicago
jamesrobinson@uchicago.edu

Shaoda Wang
Associate Professor
Harris School of Public Policy
University of Chicago
shaoda@uchicago.edu

JOB MARKET PAPER

The Human Cost of Conservation: Evidence from Protected Areas in Africa

Abstract: The share of global land designated as protected areas (PAs) more than doubled from 7.5% in 1990 to 17.5% in 2020 in response to global biodiversity decline. I examine this expansion of PAs in twenty-first-century Africa to show how the local costs of environmental policies intended to generate global benefits can produce severe unintended consequences in weak states. Combining comprehensive PA boundary data with georeferenced conflict data in a difference-in-differences framework, I find that PA designation increases the probability of conflict by 0.6 percentage points—a 50% increase relative to the baseline. This effect is concentrated in borderlands, where state authority is often limited. Several findings suggest that conflict is driven by grievances against the state arising from the local costs of PA designation. First, I find no statistically significant effects in strong states, which can address or credibly commit to addressing such grievances. Second, the effect is driven by conflict involving the state and actors with explicit anti-state agendas, with no effects for other violent political actors. Third, conflict does not increase following designation in PAs that provide greater local autonomy through community governance or potential economic benefits from safari tourism. Finally, PA designation has no meaningful effect on environmental outcomes. Together, these findings underscore the importance of accounting for local political constraints and incentives when expanding PAs in fragile states to maximize conservation benefits while avoiding severe unintended consequences.

WORKING PAPERS

Environmental Degradation in One's Own Backyard: Who Gains and Who Loses From Sand Mining in India (with Claire Fan, Varun K, and Katie Zhang)

Abstract: In the context of sand mining in India, we study why environmentally damaging natural resource extraction can persist despite substantial costs to nearby communities. Using a difference-in-differences design, we find that sand mining causes both environmental damage and local economic expansion. Near mines, the share of land inundated during the monsoon increases by 25% and satellite-measured vegetation declines by 2.8%, while nighttime light intensity rises by 12%. We then quantify the distribution of these costs and benefits. Asset gains are concentrated among socially and politically dominant groups. A survey of nearly 8,000 households across 210 mining and nearby non-mining villages reveals that only 9.0% of households in mining villages earn income from mining, with 1% of households receiving 53.8% of total mining income. Compensation for environmental damage is rare, and household profits from mining are insufficient to cover measured environmental losses in two-thirds of mining villages. To rationalize these findings, we develop a model of communal policymaking in which each household's support for resource extraction depends on its private net gains from extraction. Giving households that earn income from mining greater influence over decisions can sustain extraction in 78% of villages, including those where aggregate local losses exceed gains. Our findings highlight a political economy mechanism through which concentrated economic benefits can sustain extraction despite widespread, uncompensated environmental damage.

The Economic Costs of Climate Change From Ecosystem Disruptions: Evidence From Forest Defoliation Outbreaks in the United States (with Greg Dwyer, Eyal Frank, and Andrew M. Liebhold)

Abstract: Climate change will affect human well-being not only through direct impacts, but also through changes in ecosystem functioning that alter the flow of local amenities and disamenities. We study one such climate–ecosystem–well-being chain in U.S. forests, focusing on the weather-dependent interaction between an invasive defoliator, the spongy moth, and a fungal pathogen that normally suppresses it. Using fine-scaled defoliation maps, gridded weather, and a long-run panel of spongy moth presence, we show that warm winters and warm, dry springs—especially when they persist over multiple years—substantially increase the likelihood and spatial extent of defoliation outbreaks. We then link these ecological shocks to nearly five decades of geocoded housing transactions and estimate that exposure to defoliation lowers nearby sale prices by at least two percent, with cumulative losses of up to about six percent for properties exposed in several consecutive years. Negative price effects remain detectable for homes located up to six kilometers from defoliated areas and are roughly twice as large when defoliation is a new shock in the local housing market. The stakes of this mechanism will grow over time, as the spongy moth currently occupies only about one-third of its climatically suitable habitat and is projected to expand into many more forests and communities. Unlike direct weather–well-being channels, this class of weather–ecosystem–well-being impacts offers additional scope for mitigation and adaptation, because managers can intervene to stabilize natural enemy interactions or to substitute for declines in ecosystem functioning. Our findings show that climate-driven disruptions to natural enemy interactions can generate sizable, spatially diffuse losses in housing wealth and highlight ecosystem management as a distinct margin for climate adaptation.

The Politics Machine: Data Centers and Grassroots Political Movements in the US (with Prabidhik KC)

Abstract: Opposition to data center development is rising across the United States and has become a salient issue ahead of the upcoming midterm elections. Most accounts of this backlash attribute it to nonpolitical concerns, such as environmental impacts, rising energy prices, anti-AI sentiment, and outright NIMBYism. We instead argue that in addition to these concerns, opposition to data center development also reflects a broader political trend: declining public trust in government and large corporations. As public trust has fallen over the past two decades and reached historic lows, data center development has provided a focal issue around which previously diffuse and partisan grievances can coalesce, mobilizing grassroots opposition to governments, public officials, and large corporations. In this paper, we provide preliminary evidence consistent with this argument. First, we show that exposure to data center development increases the monthly probability of an all-cause protest in

a county by as much as 10 percentage points. Using election data, we then provide suggestive evidence that congressional districts with greater exposure to data centers experience larger competition for candidacy in the electoral process and stronger anti-incumbent effects. We next lay out a range of analyses that we intend to conduct once data from the 2026 general election become available. Finally, we discuss the implications of our findings for business strategy and electoral politics in the United States.

On Strategic Civilian Targeting in Civil War (with Gina Eckhoff)

Abstract: Where do state forces adopt more violent strategies during civil war, and what mechanisms may explain differential strategies across space? In this paper, we study state violence against civilians during the Nepalese civil war (1996-2006), and show that the state is more likely to victimize civilians in areas that are more remote from developed infrastructure and the state's presence. We use the unexpected death of the Nepalese monarch and the subsequent accession of a more violence-prone king as an exogenous shock to the timing of military deployments. We find that remote villages face a seven-fold increase in violence against civilians in the first years after the event, compared to non-remote villages. We connect our findings to differential political costs of violence in remote locations, reflecting weaker accountability and/or concern for loss of legitimacy in remote locations. Our findings have implications for what conditions may discipline state forces to behave better when faced with the opportunity to misuse power.

WORK IN PROGRESS

When Do Incentives for Preemptive Action Erode Gains from Policy? Studying the Effectiveness of the Endangered Species Act (with Eyal Frank and Anouch Missirian)

The Political Economy of Protected Area Designation

PRE-PHD PUBLICATIONS

Paudel, S. R., **Banjara, S. P.**, Choi, O. K., Park, K. Y., Kim, Y. M., & Lee, J. W. (2017). Pretreatment of agricultural biomass for anaerobic digestion: Current state and challenges. Bioresource Technology, 245, 1194-1205

Paudel, S. R., **Banjara, S. P.**, Wagle, A., & Freund, F. T. (2018). Earthquake chemical precursors in groundwater: a review. Journal of Seismology, 1-22

AWARDS AND FELLOWSHIPS

Innovations for Poverty Action 2025, **$82,868** (with Corey Vernot and Arjun Kharel) – "Increasing Legal Access for Migrant Victims of Fraud: Evidence from Nepal"

The Weiss Fund for Research in Development Economics, 2025 **$49,000** (with Claire Fan and Varun K) – "The Environmental and Human Costs of Sand Mining"

EPIC India Research Fund 2023, **$10,000** (with Claire Fan and Varun K) – "The Environmental and Human Costs of Sand Mining"

The Pearson Institute Grant, **$10,000** (with Claire Fan and Varun K) – "The Environmental and Human Costs of Sand Mining"

Development Economics Research Fund 2023, **$5,000** – "Political Economy of Construction Minerals Extraction: Evidence from Nepal"

Griffin Incubator 2021 – The Data Revolution, **$40,000** (with Eyal Frank and Anouch Missirian) – "Land-Use Restrictions and Perverse Incentives: Using Machine Learning to Evaluate the Endangered Species Act from High-Frequency Satellite Data"

International House Non-Resident Graduate Fellowship 2022 - 2025, **$3,000 per year**

TEACHING EXPERIENCE

University of Chicago (Master's level courses)

Coding Camp, Instructor	2026
Methods of Data Collection: Experiments and Surveys (PPHA 44900 1), Head TA	2026
Math Camp, Instructor	2024 - 2025
Statistics for Data Analysis I (PPHA 31002), Head TA	2022 - 2024
Data and Programming for Public Policy I - R (PPHA 30535), Head TA	2023

Yale University (Undergraduate level courses)

Intro to Computing for Scientists and Engineers in C++ (ENAS 130)	2018
Ordinary and Partial Differential Equations (ENAS 194)	2018

PROFESSIONAL EXPERIENCE

The University of Chicago

Part-time research assistant for Professor Anne Karing	2022
Part-time research assistant for Professor Eduardo Montero	2022

Energy Policy Institute at The University of Chicago

Predoctoral Fellow for Professor Eyal Frank	2019 - 2021

CONFERENCES AND WORKSHOPS

CU Environmental & Resource Economics Workshop	2026
Midwest International Economic Development Conference, UIUC	2025
Interdisciplinary Ph.D. Workshop in Sustainable Development, Columbia University	2025
LSE Environment Camp, London School of Economics	2025

SERVICE

Reviewer for Midwest International Economic Development Conference	2023
Co-organizer of the Energy and Environment Economics Weekly Student Seminar	2023
Panel Moderator for South Asia Forum - Natural Disasters Risk Management	2023
Predoctoral fellow recruitment for Energy Policy Institute at The University of Chicago	2020
Board Member for South Asian Graduate Students Association at Yale	2017 – 2019

SKILLS

Technical

R, R-spatial, Python, Stata, Google Earth Engine, Remote Sensing, Machine Learning

Languages

Nepali (Native), English, Hindi, Maithili